

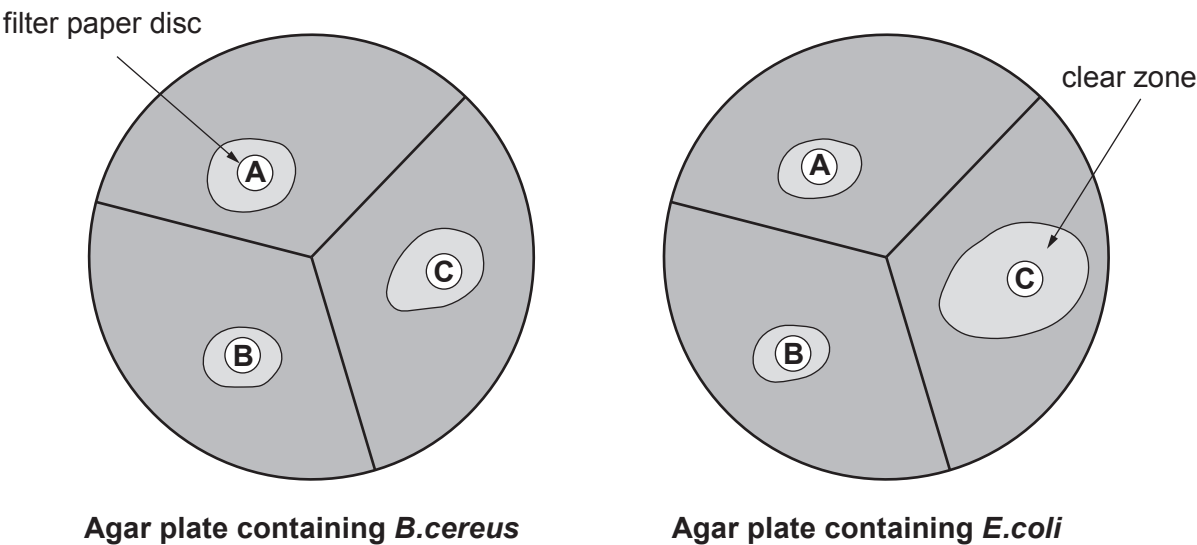
6. A class of students compared the effects of two antibiotics and an extract from the seeds of moringa plants on the growth of bacteria. Each group of students was given two agar plates. One plate had bacterium *B.cereus* growing on it whilst the other had the bacterium *E.coli* growing on it. The students used aseptic techniques to place three filter paper discs labelled **A**, **B** and **C** on each plate. Each disc had each been soaked in one of the following solutions as shown in **Table 6.1**.

Table 6.1

Filter paper disc	Solution
A	moringa seed extract
B	penicillin (an antibiotic)
C	tetracycline (an antibiotic)

The plates were incubated for 24 hours. The results are shown in **Image 6.2**.

Image 6.2



The students measured the diameter of the clear zones and recorded them in **Table 6.3**.

Table 6.3

Bacterium	Diameter of clear zone (mm)		
	Disc A moringa seed extract	Disc B penicillin	Disc C tetracycline
<i>B.cereus</i>	11	8	12
<i>E.coli</i>	9	8	18



Examiner
only

- (a) (i) State **two** steps the students would have used to maintain aseptic conditions whilst placing the discs on each plate. [2]

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- (ii) State a suitable temperature the students would have used to incubate the agar plates in a school laboratory. [1]

..... °C

- (b) (i) Explain how the clear zones formed on the agar plates shown in **Image 6.2**. [2]

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- (ii) State **two** conclusions which can be made about the effect of the moringa seed extract on the growth of bacteria compared to the two antibiotics. [2]

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- (c) Suggest **one** way in which each group of students in the class could have increased the validity of their conclusions. [1]

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